

Michael G. Nute

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EDUCATION

Ph.D., Statistics, University of Illinois at Urbana-Champaign

August 2019

- Advisor: Prof. Tandy Warnow. Dissertation: *"Statistical Estimation Problems in Phylogenomics and Applications in Microbial Ecology."*

B.A., Mathematics & Economics, Cornell University

May 2005

PROFESSIONAL & RESEARCH EXPERIENCE

Treangen Lab, Dept. of Computer Science, Rice University

Research Scientist / Postdoctoral Research Associate

Jul 2020 – Sep 2022; Sep 2023 – Present

- Lead computational and machine-learning research across a lab of 10–18 members; co-authored 20+ publications.
- Research on multiple sequence alignment (MSA), phylogeny estimation, and metagenomics: scaling Bayesian joint alignment-phylogeny estimation past 1,000 sequences, taxonomic identification for long high-error amplicon reads, novel metagenomic data-visualization, and unsupervised deep-learning models that reveal interpretable patterns in metagenomic data.
- Developed anomaly-detection and predictive methods for pathogen identification and biothreat detection; designed a biosecurity screening system for a synthetic-DNA service to meet federal requirements.
- Helped develop **Omi** (omi.rice.edu), an LLM-powered AI co-pilot on the OpenAI API for natural-language data exploration.
- Built software in one week with Claude Code and GitHub Copilot that won 2nd place (\$30K) in the DARPA Bioattribution Challenge.
- Mentored graduate and undergraduate researchers and administered lab computing resources.

Anvil Diagnostics, Southborough, MA

Co-Founder

Sep 2022 – Jul 2023

- Co-founded a diagnostics startup; built the enterprise codebase from scratch, validated product-market fit through customer discovery, and authored an SBIR grant application.

University of Illinois at Urbana-Champaign

Graduate Research Assistant & Instructor

Aug 2012 – Aug 2019

- PhD research in statistical methodology and algorithm development for large-scale sequence data, including phylogenetics, comparative genomics, and multiple sequence alignment.
- Taught three undergraduate statistics courses as TA and two as instructor of record.

Caterpillar, Champaign, IL

Graduate Research Assistant

May 2014 – May 2015

- Designed a database schema and built an ETL pipeline ingesting streaming remote-sensor data into a real-time analytics dashboard; improved company-wide demand forecasting over a 10-year horizon.

The Hanover Insurance Group, Worcester, MA

Actuarial Analyst; Corporate Development Manager

Mar 2007 – Aug 2012

- Multiple roles over five years, primarily in actuarial research and mergers & acquisitions; promoted twice and supervised a team of four. Led corporate capital and catastrophe-risk modeling efforts across \$2.6B in premium, among other projects.

Monitor Group, Cambridge, MA

Associate Consultant

Jun 2005 – Mar 2007

- Advised clients on pricing strategy and market analysis in the distribution and pharmaceutical sectors.

RESEARCH INTERESTS

- Multiple sequence alignment and phylogeny / species-tree estimation, including scalable Bayesian and coalescent-based methods.
- Metagenomics and microbial community profiling; taxonomic identification from long-read and short-read sequencing.
- Machine learning and deep learning for biological data; pathogen detection, biosecurity, and biothreat characterization.

AWARDS & HONORS

- Outstanding Postdoctoral Research Award, Rice University School of Engineering, 2022.
- Postdoctoral fellowship, National Library of Medicine Training Program in Bioinformatics & Data Science (Gulf Coast Consortia), 2020–2021.
- Doctoral fellowship, CompGen Initiative, UIUC, 2016–2017.
- Research fellowship, Caterpillar Corp., 2014–2015.
- Named to the “List of Teachers Ranked as Excellent by Their Students,” UIUC, Fall 2012.

PUBLICATIONS

(**) sole first author; (*) co-first or equal contributor. Author name shown in bold.

- B. Kille, M. **Nute**, V. Huang, A. Phillippy, T. Treangen. “Parsnp 2.0: Scalable Core-Genome Alignment for Massive Microbial Datasets.” *Bioinformatics* (in press), 2024.
- K. Curry, S. Soriano, M. **Nute**, S. Villapol, A. Dilthey, T. Treangen. “Microbial Community Profiling Protocol with Full-length 16S rRNA Sequences and Emu.” *Current Protocols*, 2024.
- A. Balaji, Y. Liu, M. **Nute**, B. Hu, A. Kappell, D. LeSassier, G. Godbold, K. Ternus, T. Treangen. “SeqScreen-Nano: a computational platform for rapid, in-field characterization of previously unseen pathogens.” *Proc. 14th ACM-BCB*, 2023.
- Q. Wang, M. **Nute**, T. Treangen. “Bakdrive: Identifying the Minimum Set of Bacterial Driver Species across Multiple Microbial Communities.” *ISMB*, 2023.
- N. Sapoval, A. Aghazadeh, M. **Nute**, D. Antunes, A. Balaji, R. Baraniuk, C.J. Barberan, et al. “Current progress and open challenges for applying deep learning across the biosciences.” *Nature Communications*, 2022.
- B. Kille, A. Balaji, F. Sedlazeck, M. **Nute**, T. Treangen. “Multiple Genome Alignment in the T2T Assembly Era.” *Genome Biology*, 2022.
- S. Soriano, K. Curry, S. Sadrameli, Q. Wang, M. **Nute**, E. Reeves, R. Kabir, et al. “Alterations to the gut microbiome after sport-related concussion and subconcussive impacts in a collegiate football players cohort.” *Brain, Behavior, and Immunity*, 2022.
- A. Balaji, N. Sapoval, C. Seto, R.A.L. Elworth, M. **Nute**, T. Savidge, S. Segarra, T. Treangen. “KOMB: Graph-Based Characterization of Genome Dynamics in Microbial Communities.” *Computational and Structural Biotechnology Journal*, 2022.
- K.D. Curry, Q. Wang, M.G. **Nute**, A. Tyshaieva, E. Reeves, S. Soriano, et al. “Emu: Species-Level Microbial Community Profiling for Full-Length Nanopore 16S Reads.” *Nature Methods*, 2022.
- K. Curry, M. **Nute**, T. Treangen. “It takes guts to learn: machine learning techniques for disease detection from the gut microbiome.” *Emerging Topics in Life Sciences*, 2021.
- B. Kille, Y. Liu, N. Sapoval, M. **Nute**, L. Rauchwerger, N. Amato, T.J. Treangen. “Accelerating SARS-CoV-2 low frequency variant calling on ultra deep sequencing datasets.” *20th IEEE Int’l Workshop on High Performance Computational Biology*, 2021.
- N. Shah, M. **Nute**, T. Warnow, M. Pop. “Misunderstood parameter of NCBI BLAST impacts the correctness of bioinformatics workflows.” *Bioinformatics*, 35(9), 2019.
- S. Roch, M. **Nute**, T. Warnow. “Long-branch attraction in species tree estimation: inconsistency of partitioned likelihood and topology-based summary methods.” *Systematic Biology*, 68(2), 2019.
- (**) M. **Nute**, E. Saleh, T. Warnow. “Evaluating Statistical Multiple Sequence Alignment in Comparison to Other Alignment Methods on Protein Data Sets.” *Systematic Biology*, 68(3), 2018.
- (*) K. Amato, J. Sanders, S.J. Song, M. **Nute**, J. Metcalf, L. Thompson, et al. “Evolutionary trends in host physiology outweigh dietary niche in structuring primate gut microbiomes.” *ISME Journal*, 2018.

- (**) M. **Nute**, E. Molloy, J. Chou, T. Warnow. "The Performance of Coalescent-Based Species Tree Estimation Methods under Models of Missing Data." BMC Genomics, 19(Suppl 5):286, 2018.
- (**) M. **Nute**, T. Warnow. "Scaling statistical multiple sequence alignment to large datasets." BMC Genomics, 17(Suppl 10):764 (RECOMB-CG special issue), 2016.
- (*) N. Nguyen, M. **Nute**, S. Mirarab, T. Warnow. "HIPPI: Highly accurate protein family classification with ensembles of HMMs." BMC Genomics, 17(Suppl 10):765 (RECOMB-CG special issue), 2016.
- J. Chou, A. Gupta, S. Yaduvanshi, R. Davidson, M. **Nute**, S. Mirarab, T. Warnow. "A comparative study of SVDquartets and other coalescent-based species tree estimation methods." BMC Genomics, 16(Suppl 10):S2 (RECOMB-CG special issue), 2015.

INVITED TALKS

- "Benchmarking BALi-Phy on Biological vs. Simulated Data," Phylogenomics Software Symposium on Large-Scale Multiple Sequence Alignment, Montpellier, France (Aug 2018).
- "SEPP and TIPP Tutorial," Strategies and Techniques for Analyzing Microbial Population Structure (STAMPS), Marine Biological Laboratory, Woods Hole, MA (Aug 2018, Aug 2019).
- "The Performance of Coalescent-Based Species Tree Estimation Methods under Models of Missing Data," RECOMB satellite meeting on Comparative Genomics (Oct 2017).
- "HIPPI: Highly Accurate Protein Family Classification with Ensembles of HMMs," Great Lakes Bioinformatics Conference (May 2017).
- "HIPPI: Highly Accurate Protein Family Classification with Ensembles of HMMs," UIUC Coordinated Science Lab Student Conference (Feb 2017).
- "Scaling Statistical Multiple Sequence Alignment Methods to Large Datasets," Phylogenomics Symposium and Software School, Austin, TX (Jun 2016).
- "Large-scale Multiple Sequence Alignment with PASTA and UPP," software tutorial, Phylogenomics Symposium and Software School (Jun 2016).
- "Large-scale Multiple Sequence Alignment with PASTA and UPP," invited tutorial for Patricia Babbitt's lab, University of California, San Francisco (Jan 2016).

SELECTED POSTERS

- "Unique Patterns of Genetic Mutation in *C. difficile* Hypervirulent Strain RT027," 31st Keck Annual Research Conference, Houston, TX (Oct 2021).
- "Mining Clinical *C. difficile* Isolates for Genomic Markers of Infection Progression," National Library of Medicine Informatics Training Conference (virtual, Jun 2021).
- "Taxon Labelling for Unidentified OTUs with Visual Phylogenetic Placement," Mid-Atlantic Microbiome Meetup, College Park, MD (Nov 2016).
- "Boosting Bayesian MCMC Multiple Sequence Alignment with Phylogeny-Aware Divide-and-Conquer," Pacific Symposium on Biocomputing, Waimea, HI (Jan 2016).

FUNDED PROJECT PARTICIPATION

- National Library of Medicine Training Program in Biomedical Informatics and Data Science (T15LM007093; PI: Kavraki).
- Elucidating the rules of cooperation and resiliency in microbial communities through stochastic graph grammars (NSF 2125965; PIs: Treangen, Bhaya).
- C3.ai Data Transformation Institute COVID-19 Award.
- Harvest Variants (CDC-funded).
- DYNAMITE: Dynamics of Colonization and Infection by Multidrug-Resistant Pathogens in Immunocompromised and Critically Ill Patients (NIH 5P01AI152999-02; PI: Arias).

TEACHING & MENTORING

- Invited teaching assistant, STAMPS (Strategies and Techniques for Analyzing Microbial Population Structure), Marine Biological Laboratory, Woods Hole, MA (2018, 2019, 2022, 2024).
- Mentored graduate and undergraduate researchers at Rice, including Ehsan Saleh, Kristen Curry, Bryce Kille, Yunxi Liu, Qi Wang, Natalie Kokroko, Emma Hammel, Lauryn Tuo, and Anika Patel.

- Coach, University of Illinois Rowing Team, 2013–2014.

TECHNICAL SKILLS

Programming: Python (expert), R, SQL, C; additional experience with Rust and Java; experienced with Claude Code and GitHub Copilot for AI-assisted development

Machine Learning & AI: Machine learning, deep learning (TensorFlow), large language models (OpenAI API), statistical modeling, data visualization

Bioinformatics & Computing: Multiple sequence alignment, phylogenetics, metagenomics, pathogen detection; reproducible pipelines (Nextflow), high-performance and distributed computing